Iván Sepúlveda
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Education

University of San Francisco Graduated August 2019
Bachelor of Science - Physics Minors in Engineering & Computer Science

Experience

Aqua Finance, Inc. - Remote *February 2026 - Present*
Manager of Analytics, Counterparty Risk

- Built counterparty risk frameworks, KPIs, early-warning indicators, predictive models, and dealer scorecards to monitor credit and fraud risk across dealers and applications
- Manage and mentor a Junior Analyst, providing technical guidance, code reviews, and prioritization while leading the team's counterparty risk analytics initiatives
- Partner with Credit, Fraud, Sales, Underwriting, Product, Data Engineering, and Operations to translate business needs into scalable risk monitoring and decision-support solutions
- Develop automated analytics pipelines and reporting using SQL and Python, improving the timeliness, consistency, and scalability of portfolio risk monitoring
- Present portfolio performance, emerging risks, and actionable recommendations to senior leadership, supporting risk strategy and dealer management decisions

Solar Mosaic LLC - Remote/Oakland, CA *June 2021 - February 2026*
Sr. Risk Analyst

- Developed SQL/Gephi network analysis using IP, geographic, and business-registration data to identify risk and fraud among connected entities
- Built an XGBoost dealer offboarding model to predict each dealer's probability of becoming offboarded at different time horizons (3M, 6M, 12M)
- Implemented dynamic partner risk scorecards, daily-updating schemas, alerts, and data initiatives with Data Engineering using `DBT, Apache Airflow, AWS, Snowflake`
- Monitored risk KPIs, conducted loan-level performance analysis, and maintained feedback loops with credit policy and operations to improve portfolio credit quality

L-Egant Solutions, LLC - Irvine, CA *October 2019 - June 2021*
Lead Software Developer [Contract]

- Developed scripts and GUI software for high-frame-rate camera operation, video capture, image algebra, pixel-intensity analysis, and two-dimensional Gaussian characterization

Mechanics Bank Auto Finance - Irvine, CA *August 2019 - June 2021*
Data Analyst - Portfolio & Servicing Risk

- Applied K-means, hierarchical clustering, and Random Forest models to segment collection agents, analyze credit metrics, inform credit policy, and forecast portfolio losses

Publications

Journal Articles

[1] C. T. Chavez et al. "Measurement of 2D density profiles using a second-harmonic, dispersion interferometer." In: *Review of Scientific Instruments* 94.2 (Feb. 2023), p. 023503. ISSN: 0034-6748. URL: https://doi.org/10.1063/5.0119896.

[2] M. Nikolic et al. "Applicability of optical emission spectroscopy techniques for characterization of Ar and Ar/O_2 discharges." In: *Journal of Physics D: Applied Physics* 54.27 (Apr. 2021), p. 275203. URL: https://dx.doi.org/10.1088/1361-6463/abf61c.

Skills

- Version Control (GitHub, Bitbucket)
- DBT, AirFlow, Docker, AWS, Snowflake
- Python, SQL, R, LaTeX, Java
- Network Analysis & Gephi
- Matplotlib, Mayavi, ggplot2, Tableau
- Monte Carlo Simulations